

MITRE ATT&CK Research Internship Project Report

Intern Information

Field	Details
Intern ID	2176
Assigned Campaign	Campaign 76
Software Range	S0760 – S0770
Submission Type	Cyber Threat Intelligence Research Report
Submission Date	15 May 2026

Project Overview

This project was prepared as part of the MITRE ATT&CK Research Internship Program. The objective of this research is to analyze cyber threat campaigns, malware behavior, threat actor tactics, techniques, and procedures (TTPs), and map them with the MITRE ATT&CK framework.

The report focuses on cyber threat intelligence activities including malware analysis, phishing monitoring, ransomware research, Indicators of Compromise (IoCs), persistence mechanisms, privilege escalation, and command & control infrastructure.

Project Objectives

1. Analyze cyber campaigns and attacker behavior

This means studying how cyber attacks are performed, what attackers target, and how they operate during a campaign.

2. Study malware and adversary software

This involves researching malicious software (malware), understanding how it works, what damage it causes, and how attackers use it.

3. Understand MITRE ATT&CK techniques

This means learning different attack techniques used by hackers and mapping them with the MITRE ATT&CK framework.

4. Research persistence and privilege escalation methods

This involves studying how attackers maintain access in a system and gain higher-level permissions after initial compromise.

5. Collect Indicators of Compromise (IoCs)

This means gathering evidence of malicious activity such as suspicious IPs, domains, hashes, URLs, registry changes, and malicious files.

6. Prepare professional cyber threat intelligence documentation

This involves creating detailed reports, summaries, timelines, screenshots, and analysis documents related to cyber threats.

7. Study phishing infrastructure and ransomware operations

This means researching phishing attacks, ransomware behavior, malicious websites, email campaigns, and attacker infrastructure.

8. Support awareness and monitoring activities

This includes helping with threat monitoring, detection activities, awareness posts, and reporting suspicious cyber activities.

Campaign Research – Campaign 76

Campaign Overview

Campaign 76 is assigned under the MITRE ATT&CK internship allocation model. The campaign research focuses on analyzing adversary objectives, malware behavior, attack methodologies, threat actor operations, targeted industries, and ATT&CK technique mappings.

The campaign includes monitoring:

Campaign Includes Monitoring —

1. Malware campaigns

Monitoring malware campaigns means tracking malicious software activities, how malware spreads, what systems it targets, and how it behaves after infection.

2. Phishing operations

This involves observing fake emails, websites, and messages used by attackers to steal credentials or deliver malware.

3. Command & Control (C2) infrastructure

C2 infrastructure refers to servers or communication channels used by attackers to remotely control infected systems and send commands.

4. Credential theft activities

This means monitoring techniques used by attackers to steal usernames, passwords, tokens, banking details, or login credentials.

5. Defense evasion mechanisms

Attackers use methods to avoid detection by antivirus, firewalls, SIEM, or security tools. Monitoring these techniques helps identify hidden threats.

6. Exfiltration methods

Exfiltration means stealing or transferring sensitive data from a victim system to attacker-controlled infrastructure.

7. Threat actor behavior

This involves studying how cybercriminals or APT groups operate, including their tactics, attack patterns, targets, and tools they use.

Threat Intelligence Activities

The project includes cyber threat intelligence activities such as :

1. Monitoring Advanced Persistent Threat (APT) groups

This means tracking organized cybercriminal or state-sponsored groups and studying their attacks, targets, and malware activities.

2. Researching ransomware and exploit campaigns

This involves analyzing ransomware attacks, exploit techniques, and vulnerabilities used by attackers to compromise systems.

3. Tracking malware infrastructure and suspicious domains

This means identifying malicious servers, domains, IP addresses, and infrastructure used for phishing, malware delivery, or command & control.

4. Collecting Indicators of Compromise (IoCs)

This includes gathering evidence such as malicious IPs, hashes, suspicious URLs, registry changes, and infected files to identify attacks.

5. Studying TTPs mapped with the MITRE ATT&CK framework

TTPs (Tactics, Techniques, and Procedures) explain how attackers operate. These techniques are mapped using the MITRE ATT&CK framework for analysis.

6. Preparing threat intelligence summaries and reports

This involves creating professional reports containing malware analysis, attack details, IoCs, mitigation methods, and threat actor information.

7. Monitoring open-source intelligence channels

This means observing publicly available sources like security blogs, threat feeds, forums, and OSINT platforms for cyber threat information.

8. Supporting threat hunting and detection activities

This involves helping identify suspicious activities in systems and networks using logs, alerts, behavioral analysis, and monitoring tools.

Important MITRE ATT&CK Techniques

Technique ID	Technique Name
T1059	Command and Scripting Interpreter
T1566	Phishing
T1547	Boot or Logon Autostart Execution
T1027	Obfuscated Files or Information
T1003	OS Credential Dumping
T1082	System Information Discovery
T1021	Remote Services
T1041	Exfiltration Over C2 Channel
T1053	Scheduled Task
T1071	Application Layer Protocol

Important MITRE ATT&CK Techniques —

T1059 — Command and Scripting Interpreter

This technique involves attackers using command-line interfaces and scripting tools such as PowerShell, CMD, Bash, Python, or shell scripts to execute malicious commands on a victim system.

Attackers use scripting interpreters because they are already available in operating systems and allow them to execute commands without installing additional software. Through these interpreters, attackers can download malware, execute payloads, create users, steal data, disable security tools, or communicate with Command & Control (C2) servers.

Example:

- Using PowerShell to download malicious files
- Running CMD commands for system control
- Executing scripts silently in the background

Risks:

- Remote system compromise

- Malware execution
- Security bypass
- Data theft

Detection:

- Monitor PowerShell logs
 - Detect suspicious command execution
 - Alert on encoded or obfuscated scripts
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T1566 — Phishing

Phishing is a social engineering technique where attackers trick users into opening malicious emails, clicking harmful links, or downloading infected attachments.

Phishing is commonly used as the initial access method in cyber attacks. Attackers may impersonate trusted organizations, banks, companies, or employees to deceive victims.

Types of Phishing:

- Email phishing
- Spear phishing
- Malicious attachments
- Fake login pages

Example:

- Fake Microsoft login email
- Malicious PDF or ZIP attachment
- Password reset scams

Risks:

- Credential theft
- Malware infection
- Ransomware deployment

Detection:

- Email filtering
 - Suspicious attachment scanning
 - User awareness training
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T1547 — Boot or Logon Autostart Execution

This technique is used by attackers to maintain persistence on a system by automatically starting malicious programs during system boot or user login.

Attackers modify startup folders, registry run keys, scheduled tasks, or services so malware runs every time the device starts.

Common Persistence Methods:

- Registry Run Keys
- Startup Folder
- Windows Services
- Scheduled Tasks

Example:

- Malware automatically launching after reboot
- Hidden startup scripts

Risks:

- Long-term persistence
- Repeated malware execution
- Difficult malware removal

Detection:

- Monitor registry changes
- Detect suspicious startup entries
- Audit scheduled tasks

T1027 — Obfuscated Files or Information

Attackers use obfuscation to hide malicious code and avoid detection from antivirus or security systems.

Obfuscation makes files or commands difficult to read or analyze. Attackers may encode scripts, encrypt payloads, or use random characters to bypass security tools.

Obfuscation Methods:

- Base64 encoding

- Encryption
- Packed executables
- Random variable names

Example:

- Encoded PowerShell commands
- Encrypted malware payloads

Risks:

- Antivirus evasion
- Hidden malware activity
- Difficult forensic analysis

Detection:

- Detect encoded commands
 - Analyze suspicious scripts
 - Behavioral monitoring
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T1003 — OS Credential Dumping

Credential dumping involves stealing usernames, passwords, hashes, or authentication tokens from operating systems.

Attackers use credential dumping to gain unauthorized access and move laterally inside networks.

Common Targets:

- Windows LSASS memory
- SAM database
- Cached credentials

Tools Used:

- Mimikatz
- PowerShell scripts
- Custom malware

Risks:

- Account compromise

- Privilege escalation
- Network-wide attacks

Detection:

- Monitor LSASS access
 - Detect suspicious credential tools
 - Enable endpoint detection
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T1082 — System Information Discovery

This technique involves collecting information about the victim system such as operating system version, hostname, installed software, hardware details, and security tools.

Attackers gather this information to understand the environment before launching additional attacks.

Information Collected:

- Computer name
- OS version
- Installed applications
- Antivirus software
- Hardware details

Risks:

- Target profiling
- Security bypass planning
- Attack customization

Detection:

- Monitor system enumeration commands
 - Detect unusual information gathering activity
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T1021 — Remote Services

Attackers use remote services to move between systems and remotely control infected machines.

Common remote services include Remote Desktop Protocol (RDP), SMB, SSH, and Windows Admin Shares.

Purpose:

- Lateral movement
- Remote administration
- Remote malware execution

Example:

- Using RDP for unauthorized access
- SMB file sharing abuse

Risks:

- Internal network compromise
- Data theft
- Spread of ransomware

Detection:

- Monitor remote login activity
 - Detect unusual SMB or RDP connections
 - Use MFA for remote access
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T1041 — Exfiltration Over C2 Channel

This technique involves attackers stealing data from victim systems through Command & Control communication channels.

Instead of using separate methods, attackers hide stolen data inside regular C2 traffic to avoid detection.

Data Exfiltrated:

- Credentials
- Documents
- Financial information
- Internal files

Common Channels:

- HTTP/HTTPS

- DNS
- FTP
- Encrypted traffic

Risks:

- Data breaches
- Intellectual property theft
- Confidential information exposure

Detection:

- Monitor outbound traffic
 - Detect unusual encrypted communication
 - Analyze large outbound data transfers
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T1053 — Scheduled Task

Attackers create scheduled tasks to execute malware automatically at specific times or intervals.

Scheduled tasks help attackers maintain persistence and repeatedly execute malicious payloads.

Uses:

- Malware persistence
- Payload execution
- Automated attacks

Example:

- Running malware every startup
- Launching scripts every hour

Risks:

- Continuous malicious activity
- Stealthy persistence

Detection:

- Audit scheduled task creation
- Monitor suspicious task names

- Alert on unknown scripts
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T1071 — Application Layer Protocol

Attackers use common application protocols such as HTTP, HTTPS, DNS, or SMTP for Command & Control communication.

Because these protocols are commonly used in normal internet traffic, malicious communication may remain hidden.

Common Protocols:

- HTTP/HTTPS
- DNS
- FTP
- SMTP

Purpose:

- Remote control
- Data exfiltration
- Malware updates

Risks:

- Hidden attacker communication
- Difficult network detection

Detection:

- Monitor network traffic anomalies
 - Analyze DNS requests
 - Inspect encrypted traffic patterns
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Software Research

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Malware Execution Methods

Malware execution methods describe how malicious software starts operating on a victim machine. Attackers use different delivery techniques depending on the target environment and security controls. Common execution methods include malicious email attachments, PowerShell commands, macros in Office documents, JavaScript payloads, DLL side-loading, and executable files disguised as trusted software.

Attackers prefer trusted system tools because they reduce suspicion and help bypass security monitoring. Once executed, malware may download additional payloads, create backdoors, establish persistence, or connect to Command & Control servers.

Example:

- A phishing email contains a malicious Word document.
- The victim enables macros.
- PowerShell downloads malware from a remote server.

Persistence Mechanisms

Persistence ensures malware remains active even after system reboot or user logout. Attackers achieve persistence by modifying startup locations, registry keys, services, WMI subscriptions, scheduled tasks, or startup folders.

Persistence mechanisms are critical because they allow attackers to maintain long-term access for espionage, ransomware deployment, or credential theft.

Common Methods:

- Registry Run Keys
- Windows Services
- Scheduled Tasks
- Startup Folder
- WMI Event Subscription

Defense Evasion

Defense evasion techniques help malware avoid antivirus, EDR, sandbox detection, and forensic analysis. Malware may encrypt payloads, obfuscate scripts, inject into legitimate processes, or disable security tools.

Attackers constantly update evasion techniques to bypass modern security systems.

Common Evasion Techniques:

- Base64 encoding
- DLL injection
- Fileless malware
- Timestomping
- Sandbox detection

Credential Access

Credential access techniques are used to steal usernames, passwords, browser cookies, authentication tokens, and session data.

Attackers use stolen credentials to move laterally, escalate privileges, and access sensitive systems.

Credential Theft Methods:

- Keylogging
- Browser password theft
- LSASS dumping
- Phishing pages
- Token theft

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Malware Capability Analysis

This involves studying the complete functionality of malware. Researchers identify what the malware can do once it infects a system.

Capabilities may include:

- Remote command execution
- Screenshot capture
- File encryption

- Keylogging
- Data exfiltration
- Webcam or microphone access

The analysis helps determine attacker objectives and potential impact.

Threat Actor Association

Threat actor association identifies which hacker group or APT organization uses the malware.

Researchers compare:

- Infrastructure patterns
- Coding style
- C2 domains
- Attack timing
- Malware behavior
- Historical campaigns

This helps attribute attacks to groups such as:

- APT29
- FIN7
- APT10

Exfiltration Methods

Exfiltration refers to how attackers steal sensitive data from victim systems.

Attackers may exfiltrate:

- Documents
- Credentials
- Databases
- Financial records
- Internal communications

Common Exfiltration Channels:

- HTTPS traffic
- DNS tunneling
- Cloud storage
- FTP
- Email attachments

Attack Chain Mapping

Attack chain mapping studies each stage of an attack from initial infection to final impact.

Typical Stages:

1. Initial Access
 2. Execution
 3. Persistence
 4. Privilege Escalation
 5. Defense Evasion
 6. Credential Access
 7. Discovery
 8. Lateral Movement
 9. Exfiltration
 10. Impact
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Threat Actor Operations

Threat actor operations refer to the complete activities performed by attackers during cyber campaigns.

These operations may involve:

- Reconnaissance
- Malware deployment
- Persistence
- Internal network movement
- Data theft
- Ransomware deployment

Researchers analyze attacker objectives, targeting methods, and infrastructure.

Privilege Escalation

Privilege escalation occurs when attackers gain higher-level permissions after initial access.

Attackers target administrator or SYSTEM privileges to gain full control over systems.

Techniques:

- Exploiting vulnerabilities
- Credential theft
- Token impersonation
- UAC bypass

Discovery Techniques

Discovery techniques help attackers gather information about the environment.

Information Collected:

- User accounts
- Network shares
- Installed applications
- Security products
- Domain controllers

This information helps attackers plan future actions.

Lateral Movement

Lateral movement allows attackers to spread across systems inside the network.

Common Methods:

- RDP
- SMB
- PsExec
- Stolen credentials
- Remote PowerShell

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Fileless Malware

Fileless malware operates mainly in memory instead of saving files to disk. This makes detection much harder because traditional antivirus tools rely on file scanning.

Fileless malware often uses:

- PowerShell
- WMI
- Registry scripts
- In-memory payloads

Encoded Commands

Encoded commands hide malicious scripts from security tools and analysts.

Attackers commonly use:

- Base64 encoding
- XOR encryption
- Custom obfuscation

Encoded commands are frequently seen in PowerShell attacks.

Alternate Data Streams (ADS)

ADS is an NTFS feature that allows attackers to hide malicious content inside legitimate files.

Security tools may miss hidden ADS content if not properly configured.

Detection Opportunities

Researchers identify behavioral indicators that security teams can monitor.

Detection Examples:

- Suspicious PowerShell activity
 - Encoded command execution
 - Memory-only malware behavior
 - Abnormal registry changes
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Browser Credential Theft

Modern browsers store passwords, cookies, and session tokens. Malware targets these stored credentials to hijack accounts.

Attackers may access:

- Banking accounts
- Social media
- Corporate VPNs
- Email accounts

Keylogging

Keyloggers record every keyboard input entered by the victim.

Captured information may include:

- Passwords
- Credit card numbers
- Chat messages
- Authentication codes

System Enumeration

System enumeration collects information about:

- Hostname
- Operating system
- Installed software
- Running processes
- Security tools

This helps attackers understand the environment.

Network Discovery

Network discovery identifies:

- IP addresses
- Active hosts
- Shared folders
- Connected devices
- Internal services

This enables lateral movement planning.

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DNS Communication

Attackers may use DNS for covert communication because DNS traffic is commonly allowed in networks.

Malware can:

- Send commands through DNS TXT records
- Exfiltrate data via DNS queries
- Resolve hidden C2 domains

Reverse Shell Activity

A reverse shell creates a remote connection from the victim system back to the attacker.

This allows attackers to:

- Execute commands
- Upload/download files
- Control the system remotely

Remote Services Abuse

Attackers misuse legitimate services like:

- RDP
- SMB
- SSH
- WinRM

This helps them move laterally without dropping additional malware.

C2 Infrastructure

Command & Control infrastructure includes:

- Domains
- Servers
- IP addresses
- Cloud services
- Proxies

Attackers use these systems to manage infected devices.

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Indicators of Compromise (IoCs)

IoCs are forensic artifacts that indicate malicious activity.

Examples:

- Malicious hashes
- IP addresses
- Domains

- Registry changes
- Scheduled tasks

IoCs help organizations detect attacks quickly.

Detection Engineering

Detection engineering involves creating detection rules for SIEM and EDR platforms.

Examples:

- Sigma rules
- YARA rules
- PowerShell alerts
- Behavioral detections

Threat Hunting

Threat hunting is the proactive search for hidden threats inside systems and networks.

Hunters analyze:

- Logs
- Network traffic
- Endpoint behavior
- Authentication activity

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Incident Response Recommendations

Incident response focuses on identifying, containing, and recovering from cyber attacks.

Main Stages:

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

SIEM Monitoring

SIEM tools collect and correlate security logs to detect suspicious activities.

Logs Monitored:

- Authentication logs
- PowerShell logs
- Firewall events
- DNS queries

Endpoint Protection

Endpoint security solutions monitor devices for malware behavior and suspicious activity.

Capabilities:

- Malware detection
 - Behavioral analysis
 - Isolation
 - Memory scanning
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Operational Threat Intelligence

Operational threat intelligence provides information about active attacks, attacker infrastructure, and emerging campaigns.

It helps organizations:

- Detect attacks early
- Understand threat trends
- Improve defenses

Threat Attribution

Threat attribution attempts to identify the group behind an attack.

Researchers analyze:

- Malware similarities
- Infrastructure reuse

- Attack timing
- Language artifacts

OSINT Collection

OSINT uses publicly available sources such as:

- Security blogs
 - Threat feeds
 - Social media
 - Leak sites
 - Research reports
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Ransomware Operations

Ransomware encrypts victim data and demands payment for decryption.

Modern ransomware groups also steal data before encryption for double extortion.

Data Theft

Attackers steal:

- Financial records
- Employee data
- Intellectual property
- Emails

Encryption Behavior

Ransomware uses encryption algorithms to lock files and prevent access.

Common algorithms:

- AES
 - RSA
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System Compromise

System compromise occurs when attackers gain unauthorized control over systems or networks.

Attackers may:

- Disable security tools
- Create backdoors
- Deploy malware
- Steal sensitive data

Attack Chain Analysis

Attack chain analysis helps researchers understand how attacks progressed step-by-step.

This analysis improves:

- Detection
- Incident response
- Security hardening

Incident Reporting

Professional incident reports include:

- Attack summary
- Timeline
- IoCs
- Impact analysis
- Mitigation steps

Detection & Mitigation

Researchers recommend security controls to prevent similar attacks in the future.

Common Recommendations:

- MFA
- EDR
- Patch management
- Network monitoring
- Threat hunting

Indicators of Compromise (IoCs)

Indicators of Compromise are evidence artifacts that help identify malicious activity in systems and networks.

Common IoCs

- Malicious IP addresses
- Suspicious domains and URLs
- Registry modifications
- PowerShell execution logs
- Unauthorized scheduled tasks
- Encoded or obfuscated scripts
- Suspicious DLL or EXE files
- Abnormal network traffic
- Credential dumping activity

Detection & Mitigation Recommendations

Detection Methods

- Enable PowerShell logging
- Monitor scheduled task creation
- Alert on suspicious registry modifications
- Monitor unusual DNS traffic
- Detect abnormal outbound traffic
- Use SIEM and EDR monitoring
- Monitor suspicious process creation

Mitigation Recommendations

- Enable Multi-Factor Authentication (MFA)
- Restrict administrative privileges
- Apply security patches regularly
- Use endpoint protection solutions
- Deploy network segmentation
- Disable unnecessary macros and scripts
- Perform continuous threat hunting

Conclusion

The MITRE ATT&CK internship project helps develop practical cybersecurity research capabilities, malware analysis skills, cyber threat intelligence knowledge, and professional reporting practices.

The project demonstrates understanding of adversary behavior, attack methodologies, Indicators of Compromise, detection opportunities, and cyber defense strategies aligned with the MITRE ATT&CK framework.

References

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